

PROJECT REPORT

EduCloud – Secure Trainer and Student File Sharing System

INTRODUCTION

EduCloud is a secure and scalable file sharing system designed to simplify the interaction between trainers and students in an educational environment. In today's digital learning era, there is a growing need for efficient platforms where study materials can be shared, assignments can be submitted, and data can be managed securely. EduCloud addresses this need by providing a centralized system that ensures reliability, accessibility, and data protection.

The project is built using a combination of Linux system administration and cloud computing services offered by Amazon Web Services (AWS). The backend infrastructure is hosted on a Linux-based EC2 instance, while services such as S3, RDS, Load Balancer, and Auto Scaling are used to enhance storage, database management, performance, and scalability. This integration allows the system to handle multiple users efficiently while maintaining high availability.

EduCloud enables trainers to upload study materials, which students can easily access and download. Additionally, students can submit assignments through the platform, which are securely stored and managed. The system incorporates multiple layers of security, including Linux file permissions, Access Control Lists (ACLs), IAM roles, and security groups, ensuring that only authorized users can access specific resources.

Overall, this project demonstrates the practical implementation of cloud-based architecture combined with Linux system management to build a real-world application. It highlights the importance of scalability, security, and automation in modern web applications and provides hands-on experience with industry-relevant tools and technologies.

OBJECTIVE

The main objective of the EduCloud project is to create a secure and scalable system where trainers can upload study materials and students can download them and submit assignments easily.

It also aims to use Linux and AWS services to ensure proper data storage, user access control, and smooth system performance.

PROJECT OVERVIEW

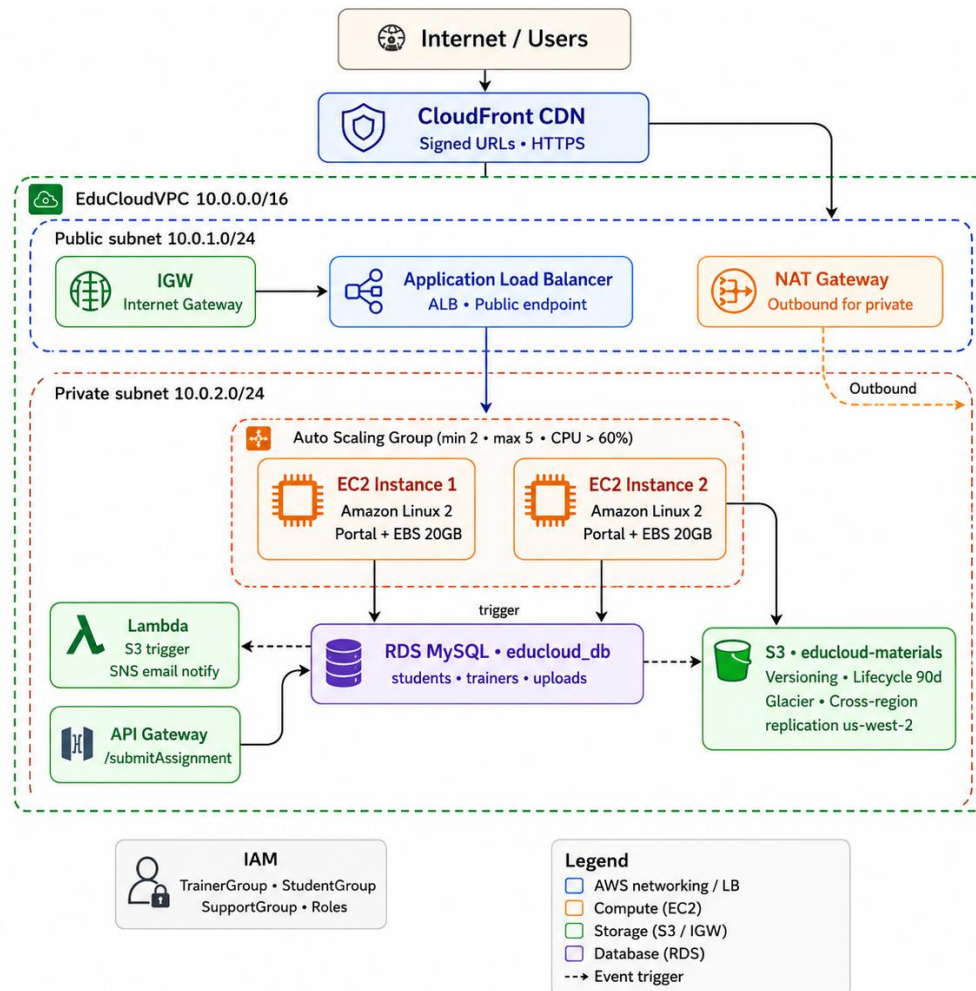
EduCloud is a web-based application developed using HTML and PHP that runs on a Linux server hosted in AWS. It provides a platform where trainers can upload study materials and students can download them and submit assignments.

The system uses AWS services like S3 for storing files, RDS for database management, and Load Balancer with Auto Scaling to handle multiple users efficiently.

Security is maintained using Linux permissions, ACL, IAM roles, and security groups, ensuring only authorized users can access the data.

Overall, EduCloud is designed to be secure, scalable, and easy to use for both trainers and students.

ARCHITECTURE DIAGRAM



User



CloudFront



Load Balancer



EC2 Server



RDS Database & S3 Storage



Response back to User

ARCHITECTURE DESIGN

A VPC is created with CIDR block 10.0.0.0/16.

Subnets created:

PublicSubnet1 – 10.0.1.0/24 (ap-south-1a)

PublicSubnet2 – 10.0.2.0/24 (ap-south-1b)

PrivateSubnet1 – 10.0.3.0/24 (ap-south-1a)

PrivateSubnet2 – 10.0.4.0/24 (ap-south-1b)

Internet Gateway is attached to VPC for public access.

NAT Gateway is created in PublicSubnet1 for private subnet internet access.

Public subnets are used for Bastion Host and Load Balancer.

Private subnets are used for EC2 application server and RDS database.

AWS SERVICES USED

VPC	– for network setup
Subnets	– for resource separation
Internet Gateway	– for public internet access
NAT Gateway	– for private internet access
EC2	– to host Linux server
S3	– to store materials and backups
IAM	– for access control
RDS	– MySQL database
ALB	– load balancing

ASG	– automatic scaling
Lambda	– serverless processing
SNS	– notifications
API Gateway	– API endpoint
CloudFront	– content delivery

SECURITY GROUPS CONFIGURATION

Security Group	Rule Type	Protocol	Port / Range	Source / Destination
BASTION-SG	Inbound	SSH	22	My IP
BASTION-SG	Outbound	All Traffic	All	0.0.0.0/0
ALB-SG	Inbound	HTTP	80	0.0.0.0/0
ALB-SG	Outbound	All Traffic	All	0.0.0.0/0
PRIVATE-APP-SG	Inbound	SSH	22	BASTION-SG
PRIVATE-APP-SG	Inbound	HTTP	80	ALB-SG
PRIVATE-APP-SG	Inbound	NFS	2049	PRIVATE-APP-SG
PRIVATE-APP-SG	Inbound	Custom TCP	20–21	BASTION-SG
PRIVATE-APP-SG	Inbound	Custom TCP	1024–1048	BASTION-SG
PRIVATE-APP-SG	Outbound	All Traffic	All	0.0.0.0/0
RDS-SG	Inbound	MySQL	3306	PRIVATE-APP-SG
RDS-SG	Outbound	All Traffic	All	0.0.0.0/0

SECURITY FEATURES

☐ **User Authentication**

Only registered users (trainer/student) can log in using username and password.

☐ **Role-Based Access Control**

Trainers can upload files, while students can only view or download files.

☐ **Secure File Storage**

Files are stored safely on cloud (AWS) to prevent unauthorized access.

☐ **Data Encryption**

Data is protected using encryption during upload and download.

☐ **Access Permissions**

Files have permissions so only authorized users can access them.

☐ **Secure Login System**

Prevents unauthorized access using proper login validation.

☐ **Network Security (AWS Security Groups)**

Controls incoming and outgoing traffic to protect the system.

☐ **Backup and Recovery**

Data can be backed up and restored if lost.

ADVANTAGES

☐ **Easy File Sharing**

Trainers can easily upload and students can access materials anytime.

☐ **Secure System**

User authentication and permissions ensure data safety.

☐ **Cloud-Based Access**

Files can be accessed from anywhere using the internet.

☐ **Scalable**

Can handle increasing number of users and files using AWS.

☐ **Centralized Storage**

All files are stored in one place, making management easy.

☐ **Time Saving**

Reduces manual sharing and speeds up communication.

LIMITATIONS

☐ **Internet Dependency**

Requires a stable internet connection to access files.

☐ **Cloud Cost**

AWS services may incur charges over time.

☐ **Basic Security Implementation**

Security features may be limited compared to advanced systems.

TESTING AND VALIDATION

Portal accessed using ALB DNS

Upload functionality tested

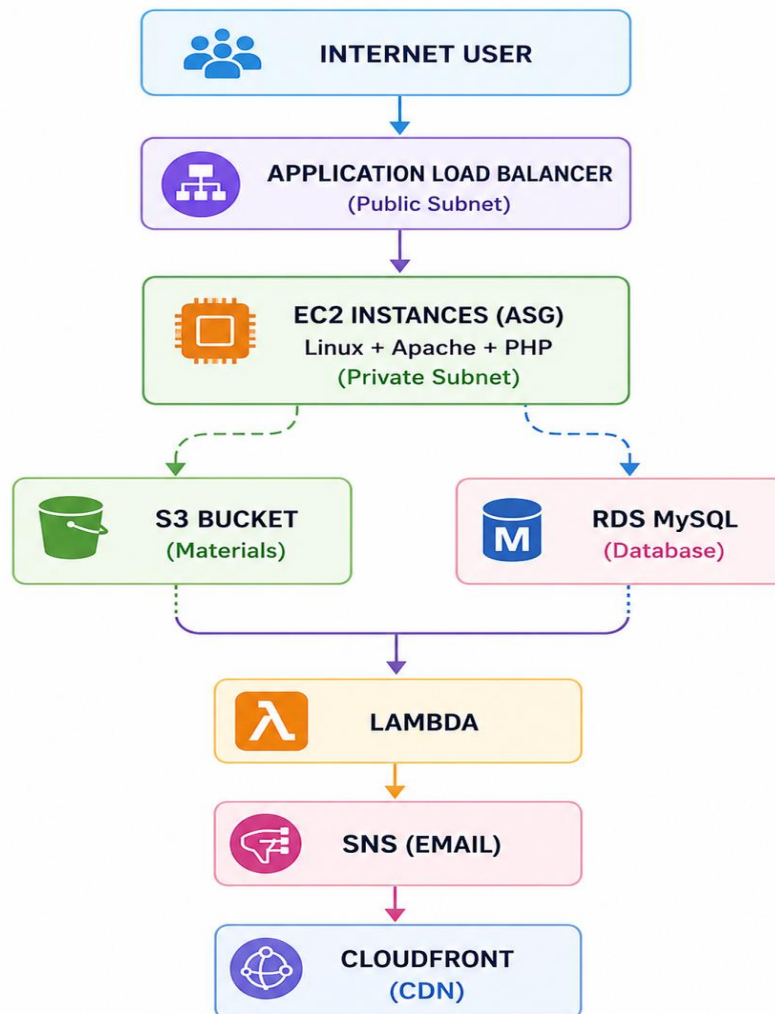
Download functionality tested

Submission functionality tested

Database entries verified

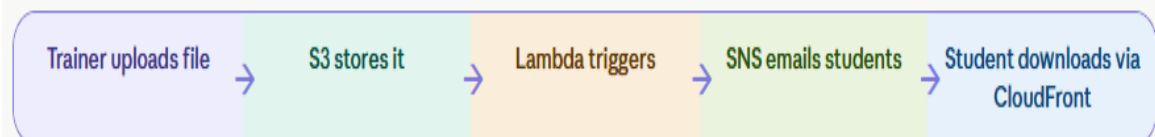
ACL permissions verified

PROJECT WORKING FLOW



User opens website --- Request goes to ALB --- ALB sends to EC2 --- Uploads go to S3 --- submissions go to RDS --- S3 triggers Lambda --- Lambda triggers SNS --- CloudFront improves delivery.

USE CASE FLOW



FUTURE SCOPE

- ❑ **Mobile Application Development**

Develop an Android/iOS app for easier access on mobile devices.

- ❑ **Live Chat or Discussion Forum**

Enable communication between trainers and students within the platform.

- ❑ **Improved User Interface (UI/UX)**

Enhance the design for a better and more user-friendly experience.

CONCLUSION

The EduCloud project provides a secure and efficient platform for trainers and students to share and access study materials. It ensures data safety through authentication and access control while allowing easy file management using cloud services. The system is simple, scalable, and helps improve the learning experience by making resources easily available anytime and anywhere.